

Institute of Engineering & Technology
BE I Year 'CSE-A & CSE-B' Class Test-II 2023
MER1C3: Elements of Mechanical Engineering

Duration: 70 Min.

Note: Attempt all questions. Use of steam table is permitted.

Max. Marks: 20

- Q1 Explain critical point and triple point in steam generation 3
- Q2 In a pressure cooker contains 2 kg of dry and saturated steam at 4 bar while in operation. Calculate the quantity of heat that must be rejected so that steam becomes wet with 0.7 dryness fraction. Also calculate the pressure and temperature of steam that exist in cooker after heat rejection. 70% 5
- Q3 Two kg of steam at 20 bar pressure exists in superheated state with temperature 250°C. Calculate 1. Enthalpy 2. Volume 3. Entropy 4. Internal energy 4
- Q4 Explain Weldability and factors of materials which affects weldability. List various welding processes. 4
- Q5 Explain Arc welding process OR gas welding process. 4